Accidents in Children

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Outline

- Definition
- Epidemiology
- Risk factors
- Prevention
- Management
  - History
  - Examination
  - Treatment

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Types

- Submersion injuries
- Choking, strangulation, and suffocation
- Poisoning
- Trauma
  - Motor vehicle accidents/Road Traffic Accidents (RTA)
  - Home injuries
  - Sports injuries
  - Burns

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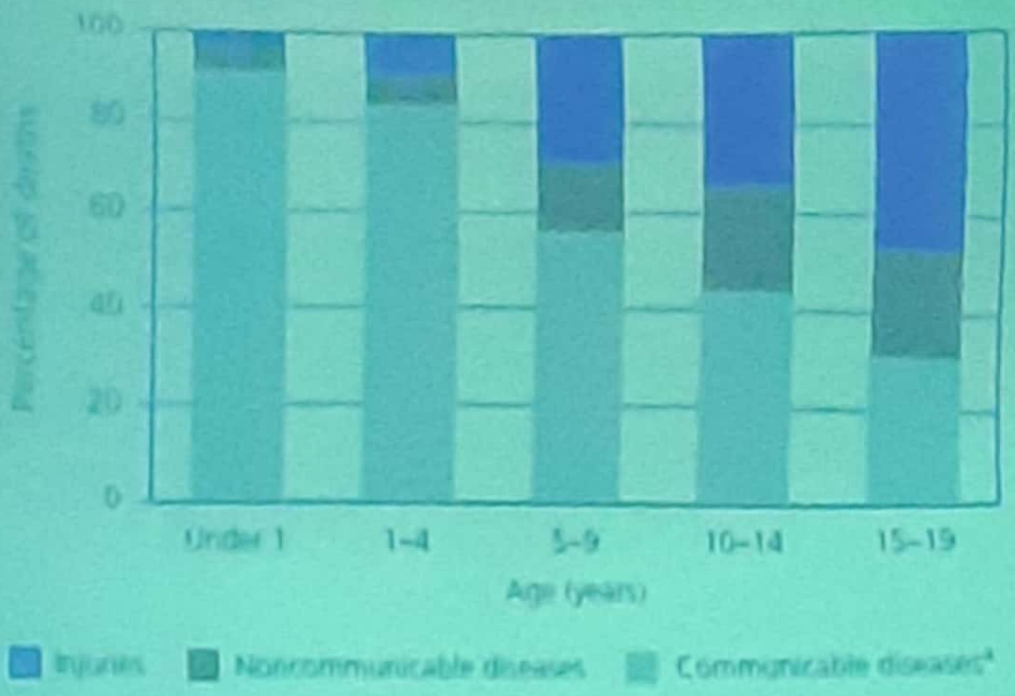
Injuries represent a large proportion of child deaths, More so for older children

Main causes of death among children, World, 2004



* Includes communicable, maternal, perinatal and nutritional conditions.

Source: WHO (2008), Global Burden of Disease: 2004 update.

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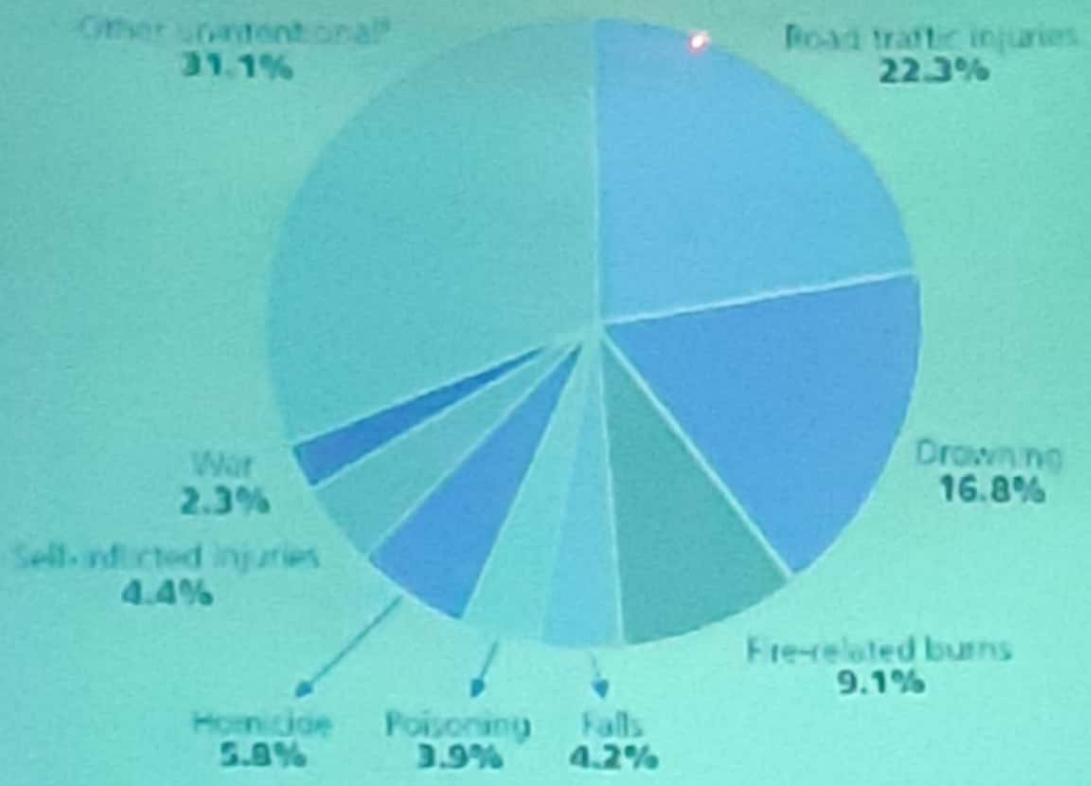
Distribution of global child injury deaths by cause, 0–17 years, World, 2004



* "Other unintentional" includes categories such as smothering, asphyxiation, choking, animal and venomous bites, hypothermia and hyperthermia as well as natural disasters.

Source: WHO (2008), Global Burden of Disease: 2004 update.

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Epidemiology

- Injuries accounted for 9% of deaths and 12% of the burden of disease worldwide in 2000.
- > 90% of global deaths from injuries occur in low- and middle-income countries.
- Males in Africa have the highest injury-related mortality rates worldwide.

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Epidemiology

- Kihiko, et al., (2010) at KNH.
  - Community-based study
    - Only a small fraction of children (4.4%) with injuries seek care at a healthcare facility
  - Commonly: cuts, scalds or burns, and falls
  - Others: animal bites, hit by moving/falling objects
  - Most of them being mild

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Epidemiology

- Kihiko, et al., 2010) at KNH in falls from a height:
  - Sustained injuries most commonly to the head and distal limbs
  - Most of the falls were from buildings and were preventable

- Gome, Mutiso & Kimende (2005) cause of trauma
  - Traumatic burns at 34.8%,
  - Falls (25.1%),
  - Foreign bodies (17.6%)
  - Road traffic accidents (8%).

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Trauma in the Community

- Most common were:
  - Cut or piercing (38.4%),
  - Fall (16.2%),
  - Burn or scald (14.3%),
  - Animal bite or kick (10.1%),
  - Hit by moving object (5.9%) and road traffic accident (3.6%).

- Of all reported injuries,
  - 30.1% sought care from traditional healers,
  - 18.4% were subject to self-care,
  - 15.4% obtained service from drug shops,
  - 4.4% were brought to a health facility for attention and
  - 17 (3.4%) took no action at all.

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Risk factors

- Single-parent households,
- Lack of parental education,
- Young maternal age
- Multiple siblings.
- Overcrowded housing,
- Lack of safe recreational facilities,
- Proximity of housing to busy streets,
- Inadequate childcare or supervision,
- Increased exposure to physical hazards
- Disabilities
  - Attention deficit hyperactivity disorder

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Falls

- Common places children fall from include:
  - Highchairs
  - Beds, changing tables, and other furniture
  - Stairs
  - Slippery floors
  - Shopping carts
  - Play equipment
  - Unsecured infant seats
  - Baby walkers
  - Windows without proper guards or stops

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Falls safety tips

- Install window guards, stair gates, and guard rails.
- Children to play in age-appropriate playgrounds with soft surfaces under the equipment.
- Never leave a baby unattended on a changing table or other piece of furniture

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Resuscitation & Stabilisation

- Prompt recognition & intervention of life-threatening emergencies

- Primary Survey
  - Airway
  - Breathing
  - **Circulation (C spine)**
  - Disability, Drugs, Depressed level of conciousness, Decontamination
  - Exposure

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Trauma

- Hypoxia and hemorrhagic shock are the final common pathways involved in pediatric trauma-related fatalities

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Management

- Assessment and management of trauma patients is divided into:
  - Primary survey and
  - Secondary survey

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Primary Survey: "ABCDE"

- Assessment and management of trauma patients is divided into the primary survey and secondary survey.
  - A=Airway (cervical spine immobilization).
  - B=Breathing.
  - C=Circulation (with hemorrhage control).
    - C spine:
  - D=Disability (a brief neurologic examination assessing the level of consciousness and pupillary size/reactivity).
  - E=Exposure (total exposure of the patient to be able to assess the entire body for possible injuries)
- Caveat: Possibility of cervical spine injury and hemorrhagic shock

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Differential Diagnosis

- High clinical index of suspicion based on the mechanism of injury
- Possibility of non-accidental trauma
  - Child abuse

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Secondary Survey

- Begins with a reassessment of the life-threatening problems addressed during the primary survey

- Followed by a complete head-to-toe physical examination to:
  - Assess and manage any non-life threatening injuries that were not identified during the primary survey.

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Prevention of RTAs

- Safer transport and land use policies
- Safety awareness in planning road networks, incorporating safety features into road design,
- Pavements (sidewalks) for pedestrians to walk separately from motorized traffic
- Produce safer vehicles
  - Improving their visibility, crash-protective designs, and the production of intelligent vehicles

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Submersion injuries

- Drowning and near-drowning
  - Bimodal age distribution
    - Infants and toddlers less than 4 years of age
      - Submersion in swimming pools, baths or household buckets
      - Lack of caregiver supervision, neglect, and suboptimal barriers
    - Adolescents
      - Associated with risk-taking behaviors
      - As well as alcohol and drug use

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Submersion injuries

- Drowning
  - Death within the first 24 hours of submersion, which includes death at the scene.
- Near-drowning
  - Submersion in which survival is greater than 24 hours, regardless of morbidity and mortality

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Submersion injuries

- Pathophysiologic events in all drowning incidents are asphyxia and hypoxia.

- Most victims will go through a period of struggle and breath-holding, or water will enter the oropharynx and larynx resulting in choking and laryngospasm

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Submersion injuries

- "Wet drowning"
  - Loss of protective reflexes will occur in most victims, leading to water aspiration.
- "Dry drowning"
  - Intense laryngospasm persists and no significant amount of water enters the lungs, even though the victim is unconscious

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Effects of submersion injuries

- All organ systems are affected after a submersion injury
  - As a result of asphyxia, hypoxia and acidosis.
- RS:
  - Respiratory failure, aspiration pneumonia, barotrauma, and adult respiratory distress syndrome (ARDS).
- CVS: Arrhythmias as well as cardiogenic shock.
- Renal dysfunction: acute tubular necrosis.
- GIT: Liver and gastrointestinal dysfunction.
- CNS: Irreversible hypoxic-ischemic damage to the brain
  - Accounts for most of the long term complications

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Management of submersion injuries

- Starts in the field with bystander cardiopulmonary resuscitation (CPR)
  - Call emergency response,
  - Be safe

- Goal:
  - Improve oxygenation and ventilation as rapidly as possible to minimize cerebral hypoxic-ischemic damage.
  - Keep warm and dry

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Safety tips

- Make sure buckets and pails are empty when you're not using them.
- Keep the toilet lid down and the bathroom door closed.
- Never leave your child unattended around water
- Barriers around water pools

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Burns

- Electrical burns and shock from inserting fingers or objects into outlets or biting electrical cords
- Flames from stoves, lamps, matches, lit cigarettes, fireplaces, and house fires
- Touching hot surfaces, such as stoves, heaters, and microwaved containers
- Hot liquid and steam from pans, cups, hot water heaters, and bath water

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Burns safety tips:

- Set water heaters no higher than 120 degrees Fahrenheit.
- Never leave food cooking unattended on the stove
- Keep pot handles turned out of reach.
- Always supervise children in the kitchen.
- Install and maintain smoke alarms on every floor of your home and near bedrooms.

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Choking, strangulation, suffocation

- Choking hazards :
  - Food, toys, batteries, bottle caps, coins, balloons, marbles, pens, buttons, rubber bands.
- Strangulation hazards
  - Necklaces, drawstrings on clothes, baby headbands, strings, ties, and ribbons as well as cords on toys, household appliances, window blinds.
- Suffocation hazards
  - Older cribs with drop rails can trap a baby, and blankets, pillows, crib bumpers, and air mattresses

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URL:

Prevention

http://www.who.int/violence_injury_prevention/child/en/

- Supervision of children
- Safety precautions such as barriers, protective gear, seat belts
- Keep dangerous items out of reach of children
- Built environment to be child friendly: walkways, guard rails
- Road safety: drivers, education

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Prevention in general

- Age appropriate toys
- Keep out of reach
  - Small objects & toys of children under three years old
  - Pull cords on curtains and blinds should be kept short
- Keep animals, especially cats, out of the bedroom and use a net on a pram
- Small food such as grapes, cherry tomatoes, blackberries and other soft fruits should cut into quarters to prevent choking.
- Tuck baby blanket in, across their chest and under their arms
- Keep the cot free from bumpers, pillows and soft toys.
- Ensure your child is the right age for the cot, bed or other sleep product
- Keep watch over children

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PREVENTION!

The most yielding role of a Paediatrician

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